

Tanuj Kumar

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[Google Scholar](#)

Education

University of Wisconsin-Madison

Fall 2021 - *

Ph.D., Department of ECE

Birla Institute of Technology & Science (BITS), Pilani, India

2015 - 2020

Integrated Dual Degree program

7.96/10

Bachelor of Engineering (Honours) in Electrical and Electronics

Master of Science (Honours) in Chemistry

Awards

1. ECE Chancellors Opportunity Fellowship (COF) award, University of Wisconsin-Madison, 2021

Publications

1. **T. Kumar**, A. Sugathan, K.L. Narasimhan, A. Pandey, S. Avasthi “CuFeSe₂ Quantum-Dot Based Infrared Photodetectors with Functionality in the Ambient” (Poster), IEEE Sensors Conference (online), Nov 2021. [\[DOI\]](#)
2. **T. Kumar**, R.E. Owyung, S. Sonkusale, “Rapid Cleanroom-Free Fabrication of Thread Based Transistors Using Three-Dimensional Stencil-Based Patterning”, Flexible and Printed Electronics, Feb 2021. [\[DOI\]](#)
3. S.M. Raje, A. Goel, S. Sharma, K. Aggarwal, D. Mantri, **T. Kumar**, “Development of On Board Computer for a Nanosatellite”, International Astronautical Congress, Sep 2017. [\[PDF\]](#)

Talks

1. **T. Kumar**, R. E. Owyung, S. Sonkusale, “High Throughput Cleanroom-Free Fabrication of Thread Based Transistors”, Materials Research Society Fall Meeting (online), Nov 2020
2. “Building a Satellite Isn’t Rocket Science”, TEDx BITS Pilani, India, Jan 2019 [\[Link\]](#)
3. “Role of Open Source in the Development of a Student-Built Nanosat & its Team Members”, Open Source Cubesat Workshop - European Space Agency, Madrid, Sep 2018 [\[Link\]](#)

Research Experience

Most closely related

1. Centre for Nano Science & Engineering, Indian Institute of Science Aug 2020 - Jul 2021
Junior Research Fellow - with Prof. Sushobhan Avasthi
 - Designed the fabrication flow and materials for ambient operation broadband quantum dot photodetectors with cleanroom fabrication and deposition techniques for improved responsivity.
 - Researched electron blocking materials and engineered bandgap to reduce dark current.

2. Nanolab, Tufts University Aug 2019 - June 2020
Visiting undergraduate student - with Prof. Sameer Sonkusale
 - Worked on fabrication of flexible, organic, ionogel-gated Thread-Based Transistors (TBT).
 - Developed a high-throughput, cleanroom-free fabrication method to increase TBT production by 100 times thereby solving manufacturing bottleneck.
 - Researched and optimized thread substrate, semiconductor and electrode materials for suitability to fabrication process.
3. Team Anant - Satellite team of BITS Pilani Jan 2016 - May 2019
Member, On-Board Computer system
 - Designed and implemented a Linux based On-Board Computer system, its drivers, and sensor interfaces for a nanosatellite to be launched into low-earth orbit.
 - Worked towards overall aim of establishing first-principles research into space systems at BITS Pilani.
 - Built the team's [website](#), learned about space electronics.

Internships

4. CentraleSupélec, Paris May 2019 - July 2019
With Prof. Caroline Lelandais-Perrault
 - Designed an asynchronous Neuromorphic ADC for use in ECG monitoring.
 - Developed a new mathematical model for its figure of merit.
 - Discovered trend of temporally decreasing maximum error in the ADC's conversion.
5. Indian Space Research Organization May 2018 - July 2018
With Dr. Vinod Kumar, FIE, FIETE, Scientific Officer
 - Designed a real-time target prediction system and its mathematical model using an Extended Kalman Filter, accurate to within 500 meters in satellite imagery.
 - Developed a software camera model as the testing framework in MATLAB.
6. Indira Gandhi Centre for Atomic Research May 2017 - July 2017
With Mr. Sanga Ramesh, Scientific Officer
 - Developed a hybrid 8-channel DMA & Interrupts based monitoring system
 - Engineered a workaround to obtain 8 inputs to system despite DAQ card supporting only 4 interrupt channels
 - Designed a GUI in LabVIEW to simultaneously view 8 channels.

Projects at BITS Pilani

7. Differential Power Attack Immune Circuits Aug 2018 - May 2019
With Prof. Anu Gupta, BITS Pilani
 - Started the Differential Power Attack project with Prof. Anu Gupta at BITS Pilani. Authored group's project proposal to the Department of Science & Technology (DST), Government of India.
 - Designed a dynamically scaled, differential pull-down circuit to increase time taken by hardware hackers.

8. Electronic properties of materials in radiation environments Jan 2019 - May 2019
With Prof. Mrinmoyee Basu
- Learned about effects of radiation typically found in the Large Hadron Collider on SiC and GaN MOSFETs to develop radiation hard semiconductors for LHC and space applications.
 - Explored effects of doping and impurities on MOSFET radiation hardness.

Teaching Assistantships

1. Microelectronic Circuits (EEE F244) Jan 2019 - May 2019
- Designed and conducted tutorial sessions on SPICE & Cadence simulation techniques.
 - Evaluated students' performances in tests.
 - Group coordinator for my TA group.
2. Computer Programming (CS F111), Aug 2018 - Dec 2018
- Conducted twice-a-week lab classes on Linux commands and C programming for freshman year students.
 - Developed problem sets and answer keys.
 - Invigilated course quizzes.

Leadership

1. Founder, Team Lead Aug 2018 - May 2019
Publicity, Sponsorship and Design subsystem - Team Anant
- Founded the subsystem to increase team's publicity, obtain sponsorships.
 - Increased publicity on social media, newspapers and through talks to increase awareness of nanosatellites and their accessibility.

Technical proficiency

1. Lab skills: Clean room fabrication, optical lithography, sputter and thermal deposition, atomic layer deposition, wet etch, Raman, FTIR & UV-Vis-NIR spectroscopy, AFM, SEM, semiconductor analysis, laser cutting, microscopy, ellipsometry
2. CAD & IDE: CleWin, Cadence tools, LTSpice, LabVIEW, EagleCAD, AutoCAD, Solidworks, Vivado, WVASE
3. Programming: MATLAB, Verilog, C, C++, Assembly, Java, Python, Git, Linux

Music

1. I dabble in the drums, guitar, digital audio work and mixing.
2. Formerly part of the Department of Music at BITS Pilani; played multiple shows.
3. Interested in progressive rock, jazz-rock, melodic metal and electronic music.

Science Communication

1. Content creator for Dr. Sabine Hossenfelder's YouTube channel "[Science without the gobbledygook](#)", where Dr. Hossenfelder separates fact from hype.
 - [Quantum Computing: Top Players 2021](#)